Experiment No-1

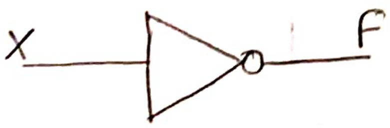
Aim:- To study & testing of various logic gates.

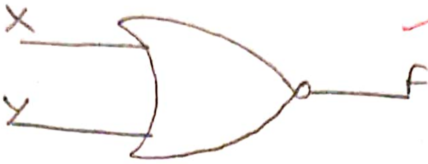
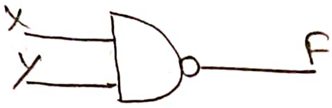
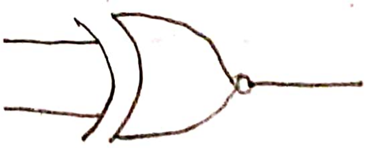
Apparatus: Digital Trainer kit. ICs, i7400, 7402, 7404, 7408, 7432, 7482, connecting wires, +5V power supply, LEDs, (or) logic probe Breadboar (if needed)

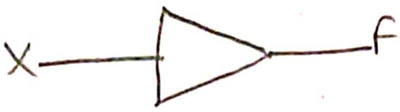
Procedure:

- Connect different gates on the digital electronics trainer kit.
- Give +5V power supply and GND, which are available on kit.
- Give the input from "Logic level inputs" section present on kit.
- Check that output from the LEDs and varify the truth table according to input and output combination.

Observation table:

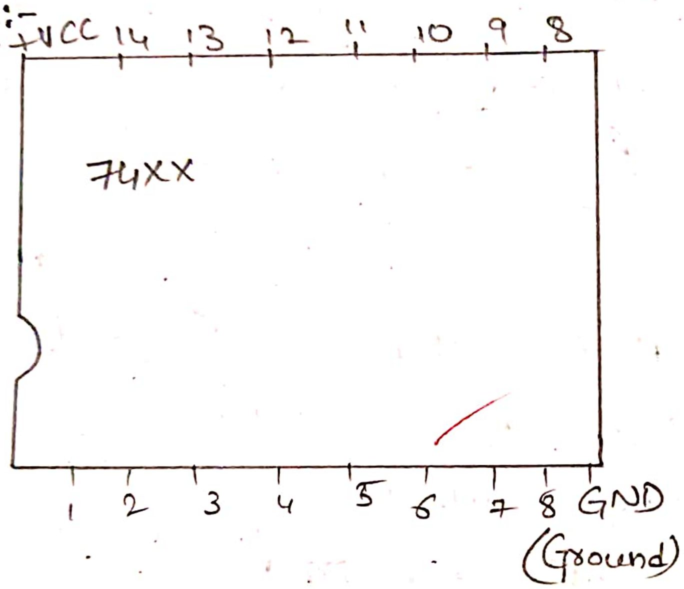
Sr No.	Name of Gate	Logic Symbol & Equation	Truth Table
1	NOT	 $F = X'$	X | F 0 | 1 1 | 0
2	OR	 $F = X+Y$	X | Y | F 0 | 0 | 0 0 | 1 | 1 1 | 0 | 1 1 | 1 | 1

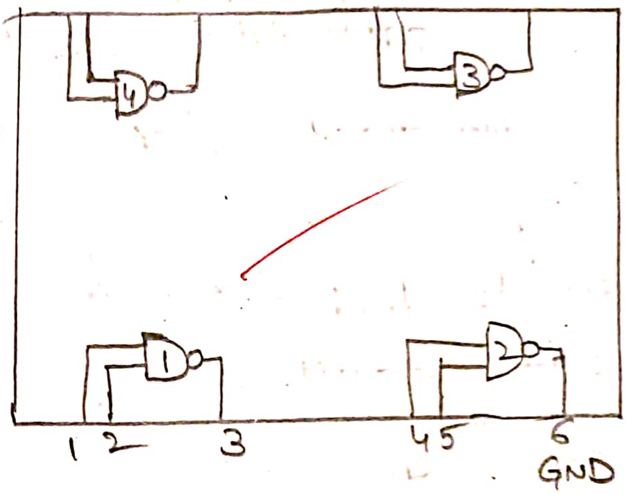
3.	AND	 $F = xy$	X Y F 0 0 0 0 1 0 1 0 0 1 1 1
4.	NOR	 $F = (x+y)'$	X Y F 0 0 1 0 1 0 1 0 0 1 1 0
5.	NAND	 $F = (xy)'$	X Y F 0 0 1 0 1 1 1 0 1 1 1 0
6.	X-OR (Exclusive OR)	 $F = x \oplus y$	X Y F 0 0 0 0 1 1 1 0 1 1 1 0
7.	X-NOR (Exclusive NOR)	 $F = x \odot y$	X Y F 0 0 1 0 1 0 1 0 0 1 1 1

8	BUFFER	 F = X	X F 0 0 1 1

* Draw a Pin diagram of above mentioned logic gates (74XX Series digital IC)

Example :-





Conclusion :-

In this experiment, we successfully studied and tested the basic logic gates : AND, OR, NOR, NAND, XOR and XNOR using a digital electronics. Trainer kit. By applying different combinations of logic level inputs and observing the corresponding outputs through LEDs we varified the function of each gate. The observed outputs matched the expected results according to their respective truth tables. confirming that the gates operate as per boolean logic. This experiment helped in understanding the fundamental working of digital circuits and rewo-forced the concept of logic gate behaviour in real hardware.

Questions:-

1. Which IC contains NAND gate?

A:- The IC 7400 Contains NAND gates

- It has four 2-input NAND gates inside.
- It is part of the 74XX digital logic IC series.
- Output is Low only when table both inputs are high.

2. How many gates are there in 7408?

A:- The IC 7408 Contains four 2-inputs-AND gates.

- It is a 14-pin IC in dual in line package -(DIP)
- Each gate gives HIGH output only when both inputs HIGH.
- It is used for performing AND logic operations.

3. Explain Modulus-2 Addition using logic gate.

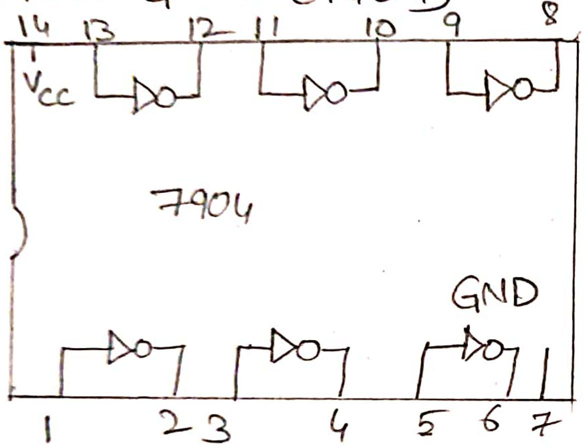
A:- Modulus-2 addition is the binary addition without carry.

- It is performed using the XOR Gate
- The output follows this rule:
  - $0 \oplus 0 = 0$
  - $0 \oplus 1 = 1$
  - $1 \oplus 0 = 1$
  - $1 \oplus 1 = 0$

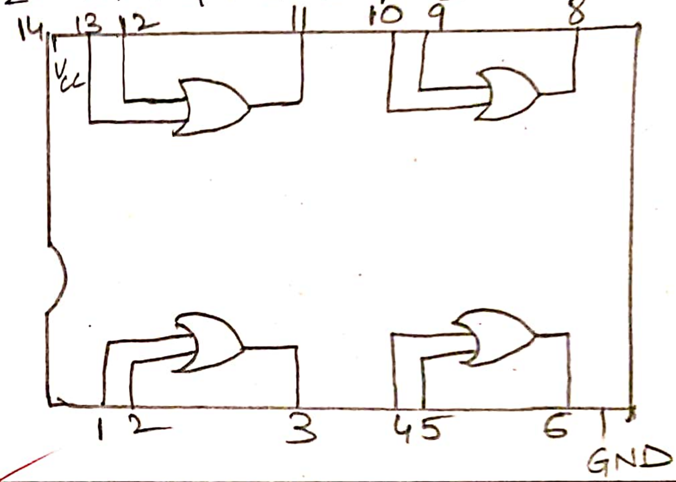
• It is widely used in binary address, parity checkers and digital circuits.

* Draw pin diagram of above mentioned logic gates 74XX Series Digital (IC):

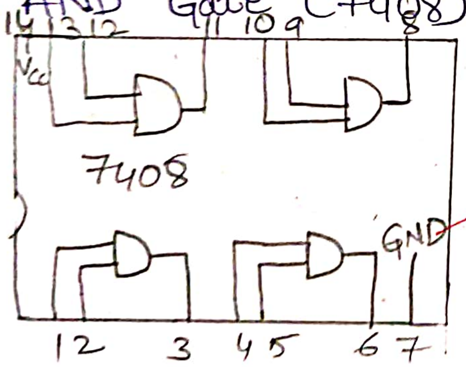
1. NOT GATE (7404)



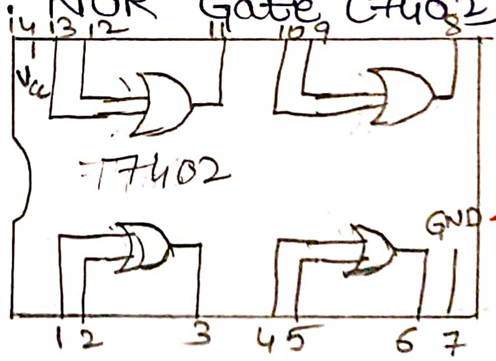
2. OR GATE (7402)



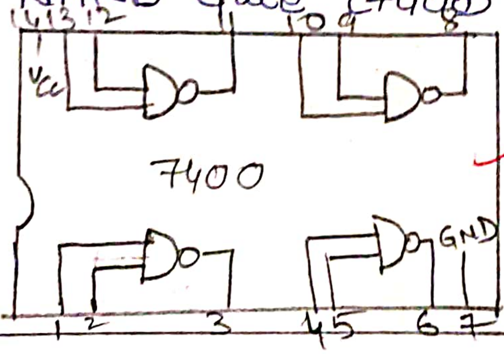
3. AND Gate (7408)



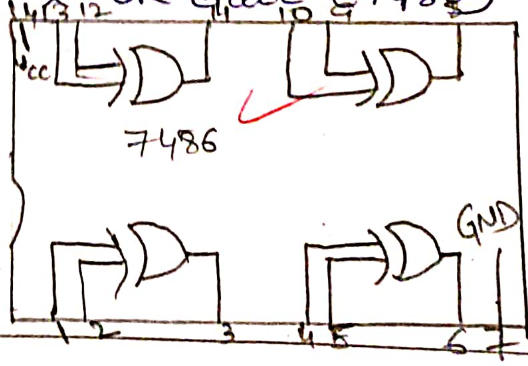
4. NOR Gate (7402)



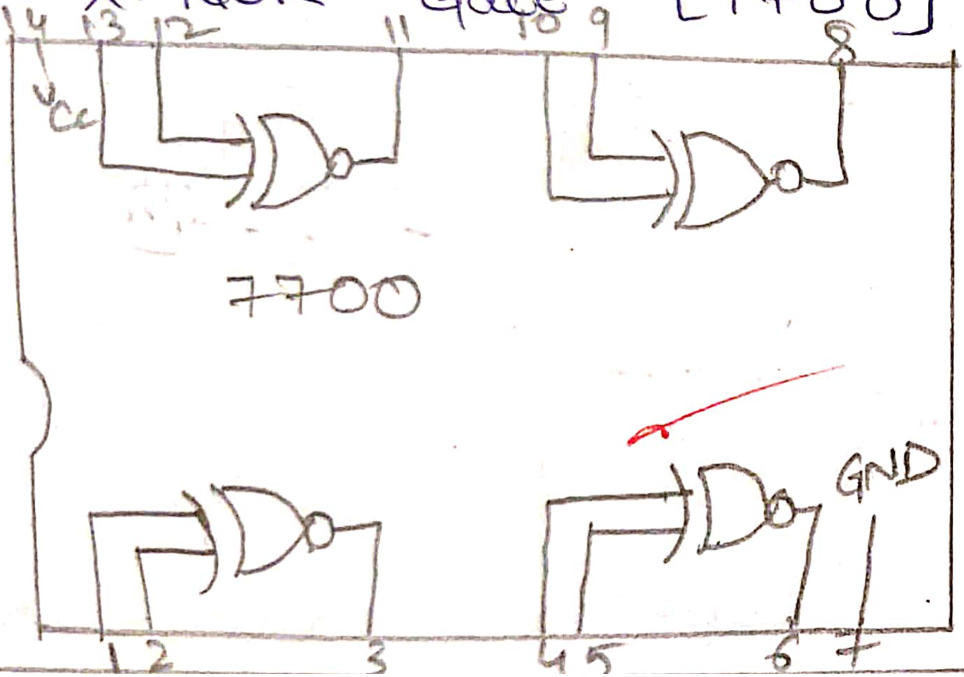
5. NAND Gate (7400)



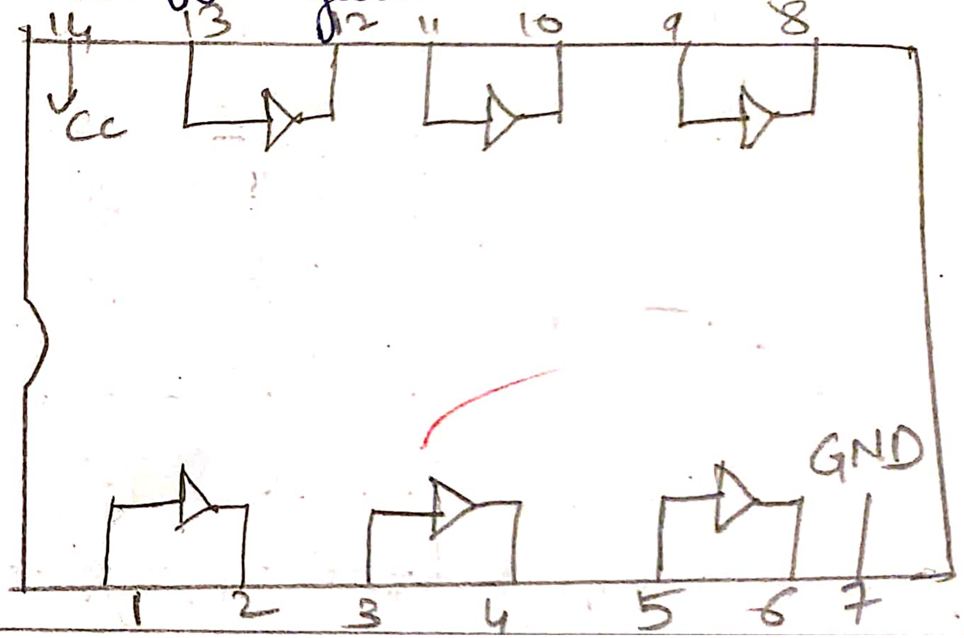
6. X-OR Gate (7486)



7. X-NOR Gate [7700]



8. Buffer gate



Experiment : 02

Aim: To study and implement all gates using following universal gates.

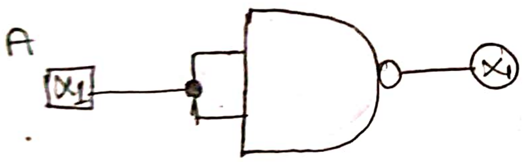
1. NAND
2. NOR

Apparatus: Digital trainer kit, patch cord (connecting wires), logic inputs switches, logic output indicates, ~~+5~~ VDC power supply, bread board (if not using trainer kits inbuilt board)
IC 7400 - Quard.

Theory: 2. Input NAND gate, IC 7402 - Quad Input NOR Gate

NAND Gate as an universal gate.

1. NOT Gate using NAND Gate

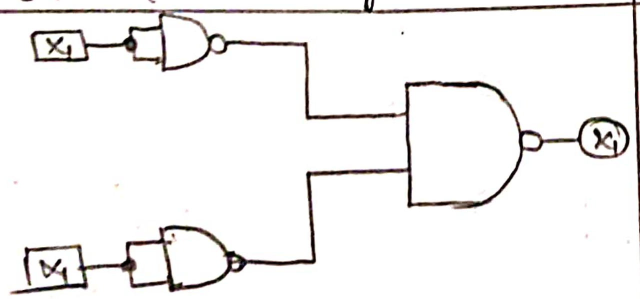


Truth Table

A	Y
0	1
1	0

NOT Gate

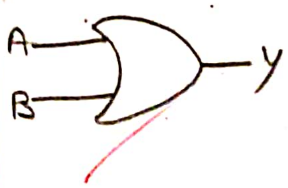
2. OR Gate using NAND Gate



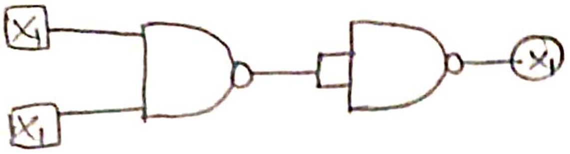
Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

OR Gate



3 AND Gate using NAND Gate



Truth table

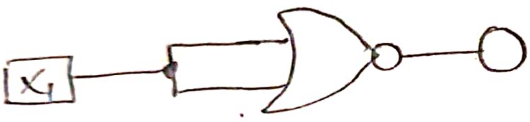
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

AND Gate

PART-B

NOR Gate as a universal gate

1 NOT Gate using NOR Gate

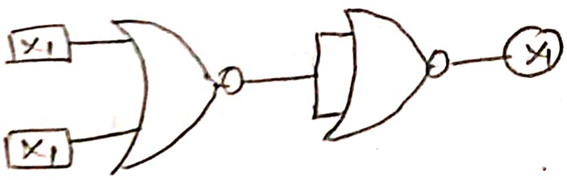


Truth table

A	Y
0	1
1	0

NOT Gate.

2 OR Gate using NOR gate

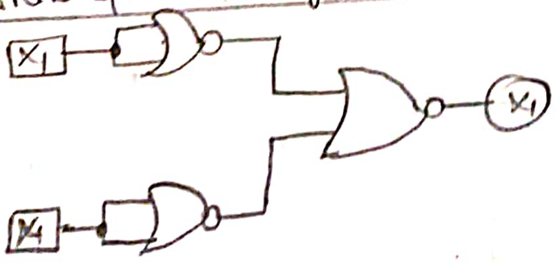


Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

OR Gate

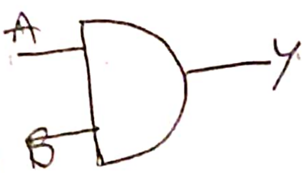
3 AND Gate using NOR Gate



Truth Table

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

AND Gate



Procedure:

1. Connect the circuit for NOT gate by using only NAND gate on trainer kit as shown in the diagram.
2. Apply different combination of input and verify the truth table.
3. fill the outputs in above blank truth table.
4. Repeat the same procedure for other gates by NAND gate and NOR gate.

Conclusion:

By Performing the experiment, we successfully implemented basic logic gates (NOT, AND, OR) using only universal gates NAND and NOR, on the digital trainer kit we connected the circuits as per the logic gate configuration using only NAND or NOR gates, applied all possible input combinations, and verified the corresponding outputs by filling in the truth tables.

This experiment demonstrates that:

- NAND and NOR gates are universal gates, meaning any logic gate (NOT, AND, OR) can be constructed using only NAND or only NOR Gates.
- The truth tables obtained from these universal gate implementations exactly matched those of the standard NOT, AND and OR Gates.
- Thus, the practical verifies the theoretical concept of university of NAND and NOR gates in digital electronics

Questions :-

1. Why NAND and NOR are called universal gate?

A:- NAND and NOR Gates are called universal gate because:

- Any basic logic gate (AND, OR, NOT) can be constructed using only NAND (or) only NOR Gate.
- They can be used to build any complex digital circuits.
- This makes them highly versatile and essential in digital electronics.

2. Design XOR Gate using NOR Gate.

To design X-OR Gate using only NOR gates the logic expression is :-

A XOR B = (A NOR (A NOR B)) NOR (B NOR (A NOR B))

This requires:

- 4 NOR Gates.
- Connect as per the logic above to get X-OR output.

logic circuit steps:-

1. Let X = A NOR B
2. Let Y = A NOR X
3. Let Z = B NOR X
4. Output = Y NOR Z

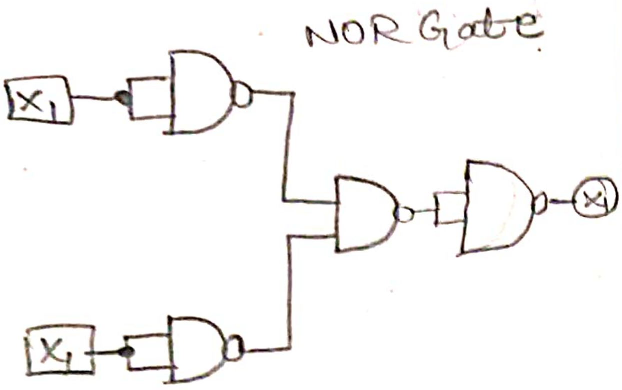
3. Which ICs are used for NAND and NOR Gates.

NAND gate IC : 7400 - (Quad 2 input NAND gate)

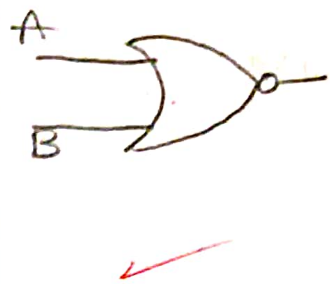
- NOR Gate IC : 7402 (Quad -2- input NOR Gate)

These are TTL based standard logic ICs used in digital circuits.

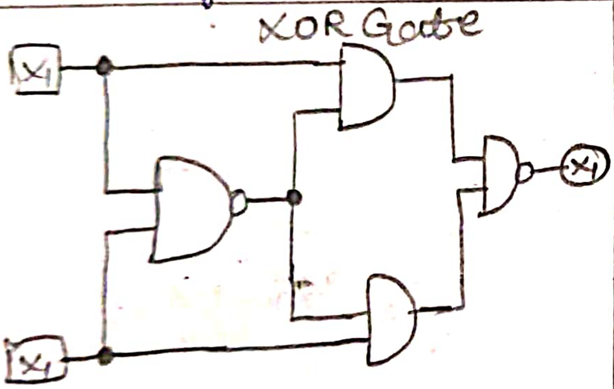
NOR Gate using NAND | Truth table | NOR Gate



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

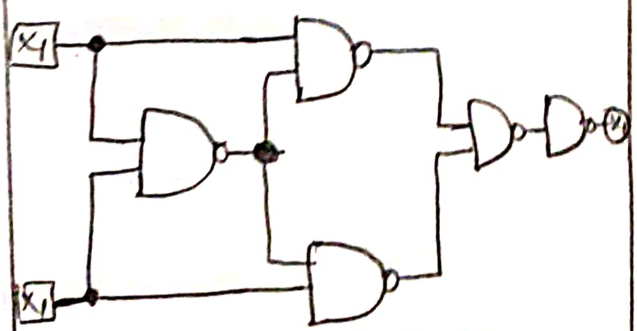


5 NOR using NAND Gate | Truth table | XOR Gate



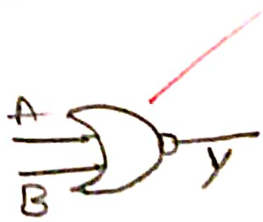
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

6. X-NOR gate using NAND Gate | Truth table | X-NOR Gate

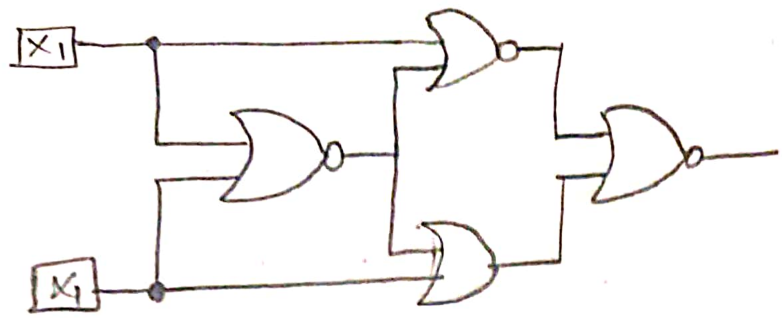


X NOR Gate

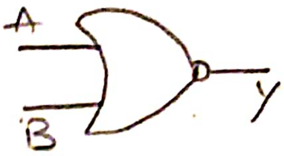
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1



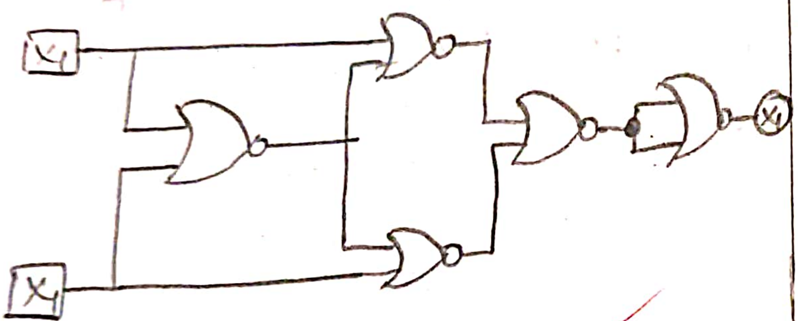
4. X NOR Gate using NOR Gate	Truthtable	X NOR Gate



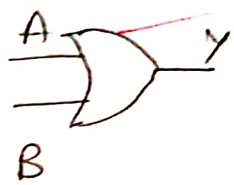
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

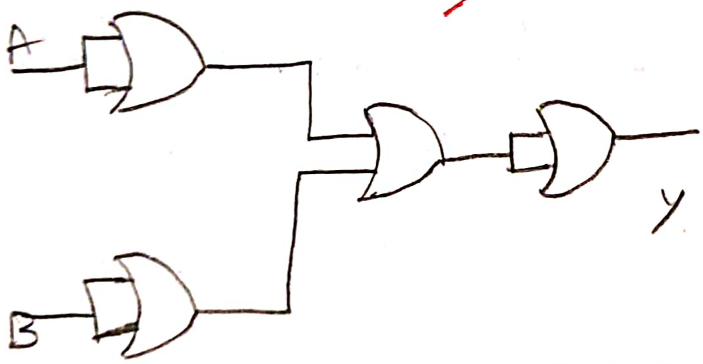


5. EXOR Gate using NOR Gate

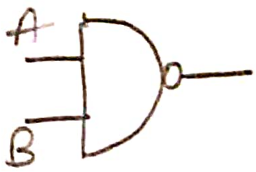


A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

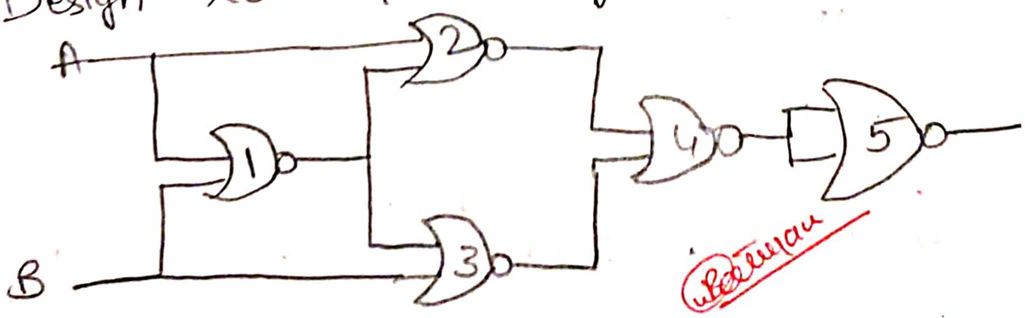




A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



Q8. Design XOR Gate using NOR Gate



$Y = \overline{A \oplus B}$

$Y = \overline{A}\overline{B} + \overline{A}$

Experiment No : 3

Aim: To study and design combinational circuits. Half adder and full adder.

Apparatus:
Digital electronics Trainer kit, powder supply, GND, LEDS

Theory:

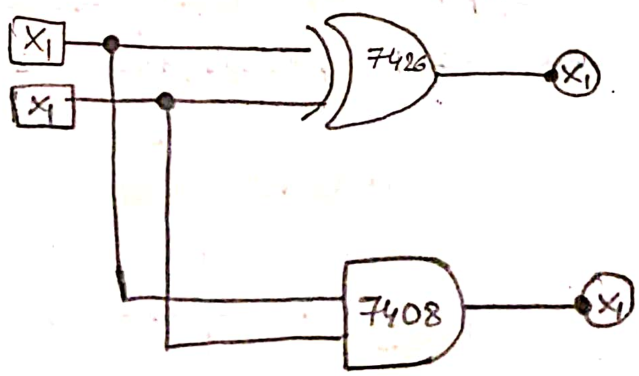
Part A : Half - ADDER

A Combinational circuit that perform the additional of two bits. is known as half adder. It requires two binary inputs and two outputs. The input variables indicate augends and added bits; the output variables indicate the sum and carry.

$S = A'B + AB'$

$C = AB$

1. logic diagrahm



Sum = AB' + A'B.
CARRY = A.B

II) Truth Table

Input		Output	
A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

PART-B: FULL ADDER

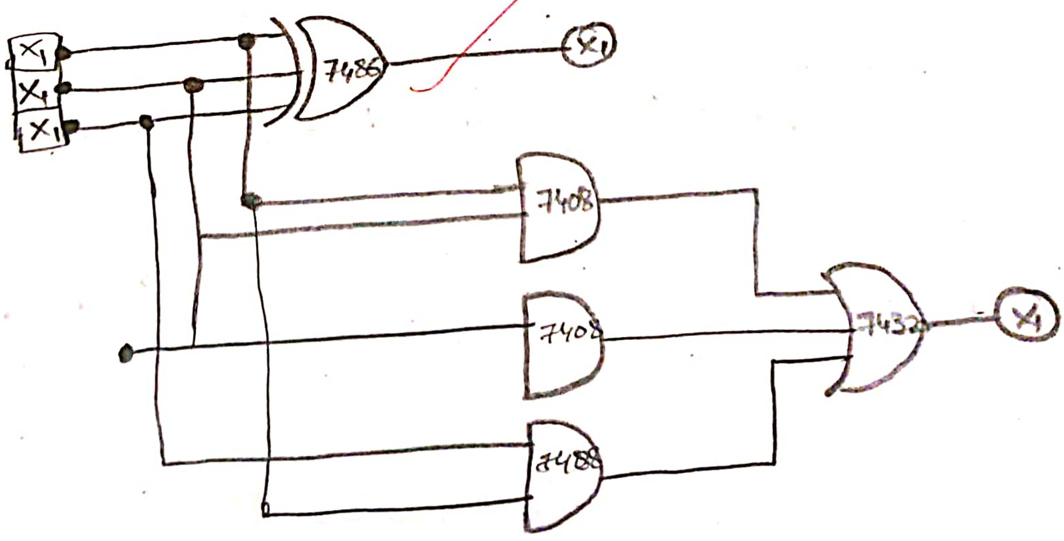
A Combinational circuit that performs that addition of two significant bits and a previous carry [three bits] is known as a full adder.

A full adder can be implemental with two half-adder it requires three binary inputs and two outputs. Two of the three input variable indicates two significant bits to be added the third input represents the carry the previous lower significant position.

(III) Logic Diagram:

$$Sum = A \oplus B \oplus C$$

$$Carry = (A \oplus B)(n-1) + AB$$



(IV) Truth Table

Input			Output	
A	B	C	SUM	CARRY
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Procedure:

- Connect. the circuit as shown in the gate diagram
- Apply different input combinations.
- See the output for different combinations and complete truth table.
- Compare practical and theoritical value from truth table.

Conclusion:

In this experiment we studied ~~amd~~ implemented combinational circuits like half adder and full adder we verified the truth table by apply different input Combinations are compared the practical outputs with theoretical values, The practical results matched the ~~expelt~~ expected outputs. proving the correct working of the circuits additionally the implementation of a full adder using NAND gate. demonstrated the concept of universal gates in digital electronics.

Questions:-

1. Define Combinational circuit

A: A Combinational circuit is a type of digital circuit where the output depends only on the current inputs, not on any previous inputs or outputs Examples includes adders multiplixers and encoders.

2. How do to convert Half adder into full adder?

A: A full adder can be built using two half adder and one OR gate:

- First half adder adds the two input list (A and B)
- Second half adder adds the two input list ⊕.
- result of the first half adder (sum) with the carry in (Cin).
- The final sum = $(A \oplus B) \oplus Cin$
- The carry = $(A \oplus B) \oplus in$
- The carry = $(A\ B) * Cin\ (A \oplus B))$

3. How many NAND gates are required to make a full adder?

A: A full adder requires a NAND gates.

- XOR operation using NAND gates : 4gates.
- AND operation using NAND gate : 2gates
- OR operation using NAND gate : 3 gates

so, the total = 4 (for XOR) + 2 (for AND) + 3 (for OR)

Sum = $A \oplus B \oplus$ carry in

Carry = $AB + (A \oplus B)$ carry is

Experiment No :- 4

Aim: To study and design combinational circuits. Half Subtractors and full subtractor.

Apparatus: Digital electronics, Trainer Kit, power supply, GND, LEDs.

Theory

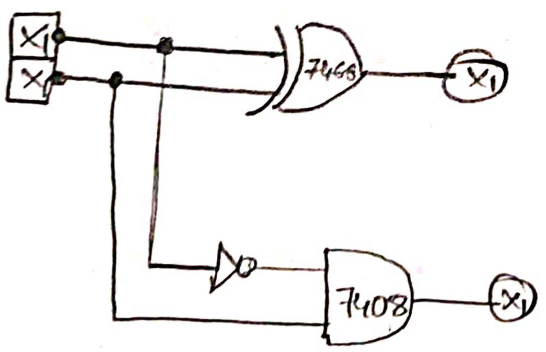
PART (A) : HALF SUBTRACTOR

A Combinational circuit that performs the subtraction of two bits is known as a half-subtractor. It requires two binary inputs and two outputs. The input variables indicates augents and added bits, the output variables indicate the difference and borrow.

$$D = A'B + AB'$$

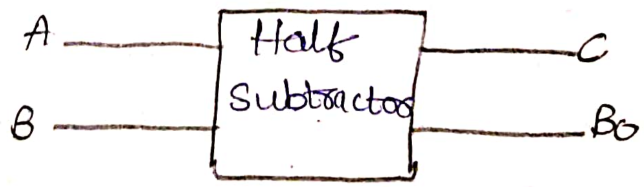
$$B = A'B$$

1. Logic Diagram:



Truth Table

A	B	D	Bo
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

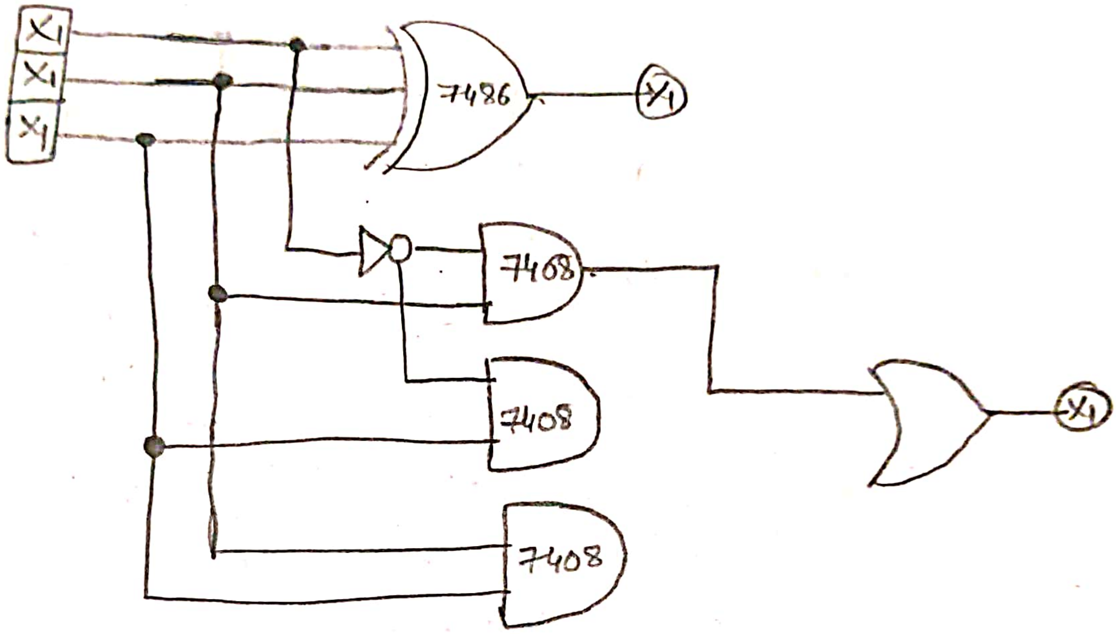


Black diagram.

PART B: FULL-SUBSTRACTOR

A Combinational circuit that performs the subtraction of the significant lets and a previous borrow [three bits] is known as Full-subtractor. A full subtractor can be implemented with two half subtractor. It requires three binary inputs and two outputs. Two of the three inputs variable indicates two significant bits to be subtracted the third input represents borrow from the previous lower significant position.

(iii) LOGIC DIAGRAM



(IV) TRUTH

INPUT			OUTPUT	
A	B	C	DUFF	BORROW
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

* Procedure:

- Connect circuit for different subtractor as shown in gate diagram.
- Apply logic voltage level to inputs of circuit from logic switches.
- See the outputs for different input combination and computes truth table.
- Compare practical and theoritical value from truth table.

Conclusion:

The comparision of practical and theoritical value from the truth table for given subtractor circuit allows for validation of the circuits functionality and understanding of any discrepancis between expected and observed behaviour.

* Questions

1. How many NOT Gate required for full subtractor?

A:- A full subtractor requires a total of 2 NOT gate when implemented using basic logic gates.

2. Construct & find equation of half Substractor using K-Map.

A:- Inputs : A.B

Output : Difference (D) Borrow (B.out)

PART B: FULL-SUBSTRACTOR

A combinational circuit that performs the subtraction of the significant b. lets and a previous

... circuits additionally the implementation of a ... using NAND gate demonstrated the concept of universal gates

- Equations
  - Difference (D):

$D = A \oplus B = A\bar{B} + \bar{A}B$

- Borrow (B-out)

$B_{out} = \bar{A}B$

for difference

A \ B	0	1	
0	0	1	$\bar{A}$
1	1	0	A
	$\bar{B}$	B	

Difference $= A\bar{B} \oplus \bar{A}B$

$= A \oplus B$

for Borrow

A \ B		1	
0	0	1	$\bar{A}$
1	0	0	A
	$\bar{B}$	B	

Borrow $= \bar{A}B$

3. Construct & find equations of full subtractor using K-Map.

A:- K-Map for full-substractor

for Difference.

A \ BB_{in}	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$D = A'B'B_{in} + AB'B_{in} + A'B_{in} + ABB_{in}$

$= A \oplus B \oplus B_{in}$

for Borrow

A \ BB_{in}	00	01	11	10
0	0	1	1	1
1	0	0	1	0

$B = A'B_{in} + A'B +$

$= \bar{A}B + (\bar{A} + B)$

PRACTICAL-5

Aim:- To design and realize a combinational circuit to convert binary to gray/gray to binary code.

Apparatus: Digital electronics, Trainer kit, power supply, GND. LEDs.

Theory:

PARTA: Binary to Gray code conversion.

The logical circuits which converts binary code to equivalent gray code is known as binary to gray code converts. The gray code is a non weighted code. The successive gray code differs in one bit position. only that means it is a unit distance code. It is also preffered as cyclic code. it is not suitable for arithmatic operation. It is most common of the unit distance codes. It is also reflexive code. An n-bit Gray code can be obtained by reflexting an n-1 bit code about an axis after 2^{n-1} rows. and putting the MSB of a above the axis and the MSB of 1 below the axis. Reflection of gray codes is shown below.

Binary code: $b_3 \oplus b_2 \oplus b_1 \oplus b_0$

Grey code: g_3 (b_3) $\quad g_2$ $(b_3 \oplus b_2)$ $\quad g_1$ $(b_2 \oplus b_1)$ $\quad g_0$ $(b_1 \oplus b_0)$.

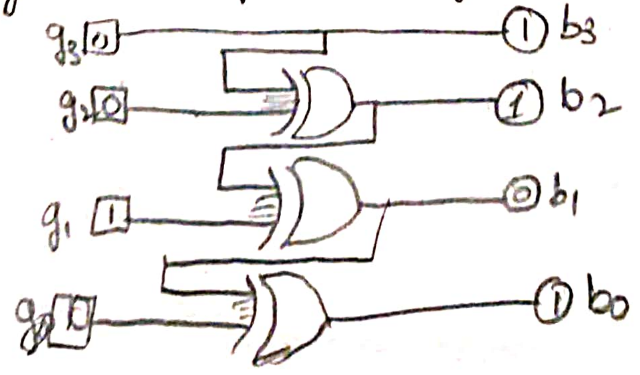
[Conversion technique]

• Equations

① Observation Table:-

Sr No.	Gray Code				Binary Code			
	G_3	G_2	G_1	G_0	B_3	B_2	B_1	B_0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0
4	0	1	0	0	0	1	0	0
5	0	1	0	1	0	1	0	0
6	0	1	1	0	0	1	1	0
7	0	1	1	1	0	1	1	0
8	1	0	0	0	1	0	0	1
9	1	0	0	1	1	0	0	0
10	1	0	1	0	1	0	1	1
11	1	0	1	1	1	0	1	1
12	1	1	0	0	1	1	0	1
13	1	1	0	1	1	1	0	0
14	1	1	1	0	1	1	1	1
15	1	1	1	1	1	1	1	0

→ Circuit Diagram [Gray to binary]



grey to binary

Procedure :-

- The circuit connections are made as shown in fig.
- Pin (14) is connected to +Vcc and Pin (7) to ground.
- In the case of lineary to grey conversion. the inputs, B_0, B_1, B_2 and B_3 are given at respective pins and outputs G_0, G_1, G_2 and G_3 are taken for all 16 combinations of the input.
- In the case of gray to binary conversion the inputs G_0, G_1, G_2 and G_3 are given at respective.
- The values of the outputs are talenbted.

Conclusion :-

The experiment successfully demonstrate Binary to gray and gray to binary conversion using k-map and verifies all possible inputs with accurate output logic.

* Questions

1. Find Equation of Binary to grey code conversion using K-map.

A: Let the binary inputs B_3, B_2, B_1, B_0 (MSB To LSB) Gray outputs are G_3, G_2, G_1, G_0.

G_0

$b_3 b_2$ \ $b_0 b_1$	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	0	1	0	1
10	0	1	0	1

for g_1

$b_3 b_2$ \ $b_1 b_0$	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	1	1	0	0
10	0	0	1	1

for g_2

$b_3 b_2$ \ $b_1 b_0$	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	1	1	1	1

Using k-map Simplification

- $G_3 = B_3$
- $G_2 = B_3 \oplus B_2$
- $G_1 = B_2 \oplus B_1$
- $G_0 = B_1 \oplus B_0$

$G[n] = B[n] \oplus B[n+1]$ (except $G_3 = B_3$ directly)

for g_3

b_3b_2 \ b_1b_0				
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

2. List out applications of code convertors:

A: used in digital Communication system to reduce transmission errors.

- In rotatary encoders to avoid multiple changes during bit transitions.
- Memory addressing and error detection.
- Dates comparision techniques.
- Interfacing different digital devices using different coding System.
- Input/Output encoding in microprocessors.

3. Find Equation of grey to binary code conversion using k-map

A: Let the Grey inputs be G_3, G_2, G_1, G_0

Binary outputs are B_3, B_2, B_1, B_0.

- $B_3 = G_3$
- $B_2 = B_3 \oplus G_2$
- $B_1 = B_2 \oplus G_1$
- $B_0 = B_1 \oplus G_0$

$B[n] = B[n-1] \oplus G[n]$

(Starting from $B_3 = G_3$)

for b_0

g_3g_2 \ g_1g_0				
00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

for b_1

g_3g_2 \ g_1g_0	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	0	0	1	1
10	1	1	0	0

for b_2

g_3g_2 \ g_1g_0	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	1	1	1	1

for b_3

g_3g_2 \ g_1g_0	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

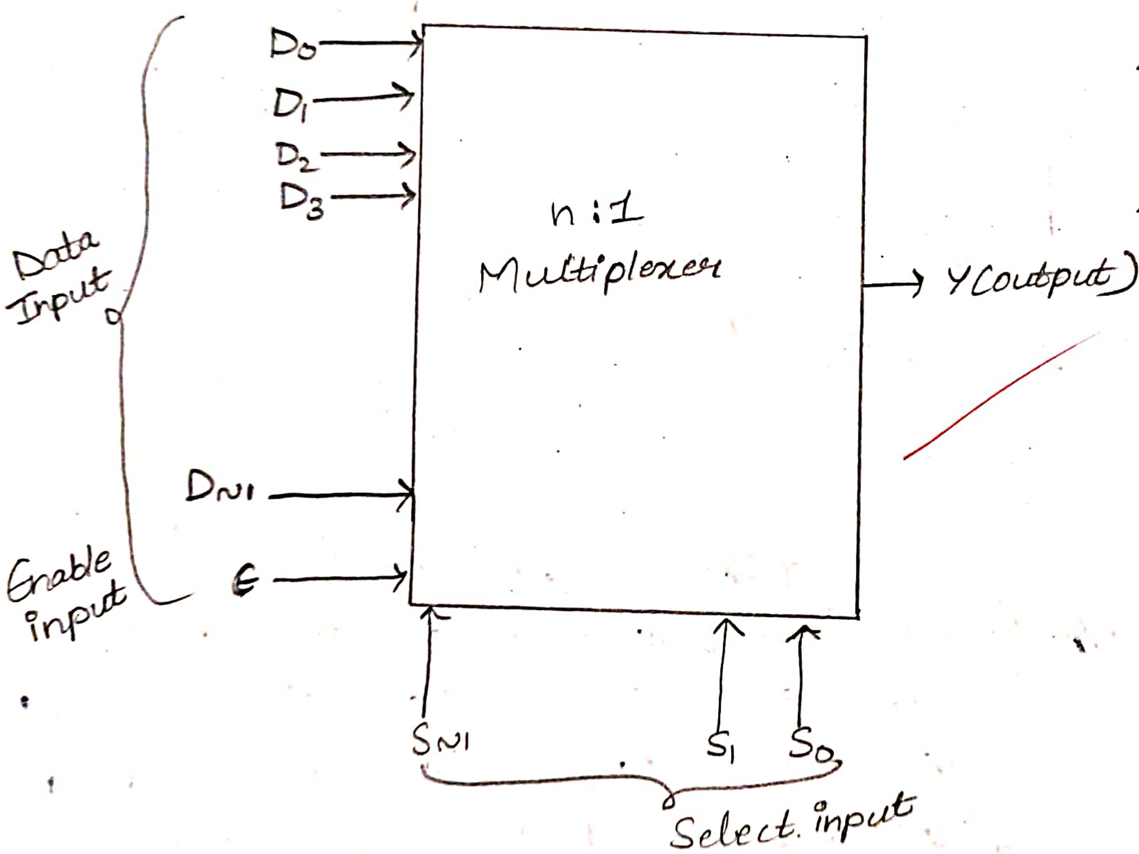
Experiment -6

Aim: To study and design of multiplexer.

Apparatus: 4 to 1 line multiplexer, kits power supply, logisim software.

Theory: A Multiplexer is a special type of combination circuit which select one of the n data inputs and routes it to the output. The selection of one of the 'n' input is done with the help of select inputs.

Block diagram:-



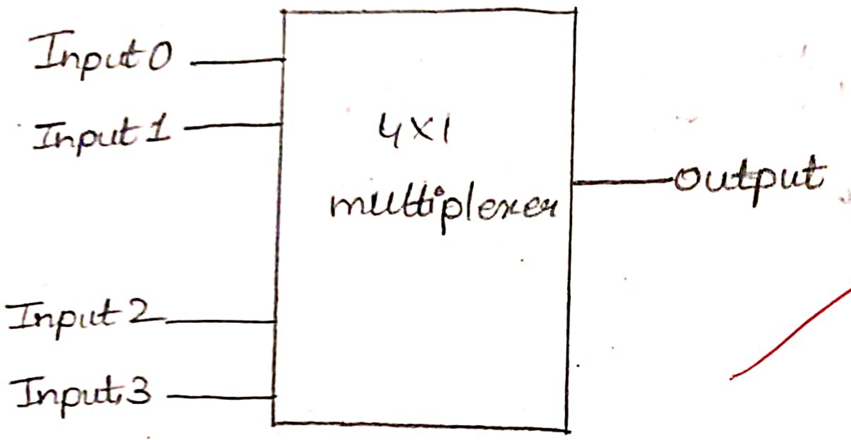
Types

1. 2*1 Multiplexer
2. 4*1 Multiplexer
3. 8*1 multiplexer
4. 16*1 multiplexer

Procedure:

- The Pin [16] is connected to +Vcc.
- Pin [8] is connected to ground.
- The inputs are applied either to 'A' input or 'B' input.
- If MUM „A" has to be initialized, Ea is made low and MUX, B" has to be initialized, Eb is made low.
- Based on the selection lines one of the inputs will be selected at the outputs and thus verify the truth table.

Logic symbol 4:1 multiplexer.



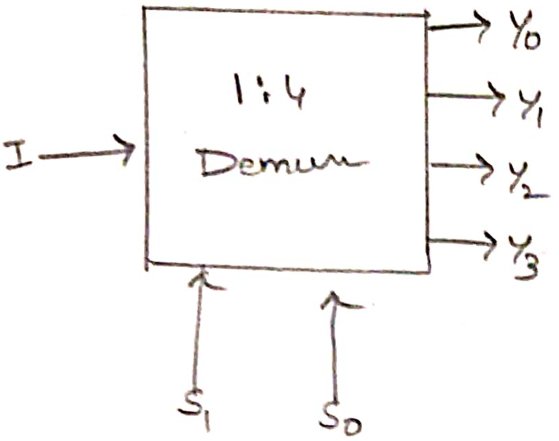
S_0	S_1	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

$$y = S_0S_1I_0 + S_0S_1I_1 + S_0SI_2 + S_0S_1I_3$$

Theory: (Demux)

A demultiplexer is a combinational logic circuit that accept a single input and distributes it over several output lines. Demultiplexer is also known as Demux is short. As Demultiplexer is used to transimit the same data to different destinations, hence it is also known as data distribute Demultiplexer is a 1 to 2ⁿ demux.

logic symbol 1:4 Demultiplexer.



$Y_0 = S_0 S_1 I$

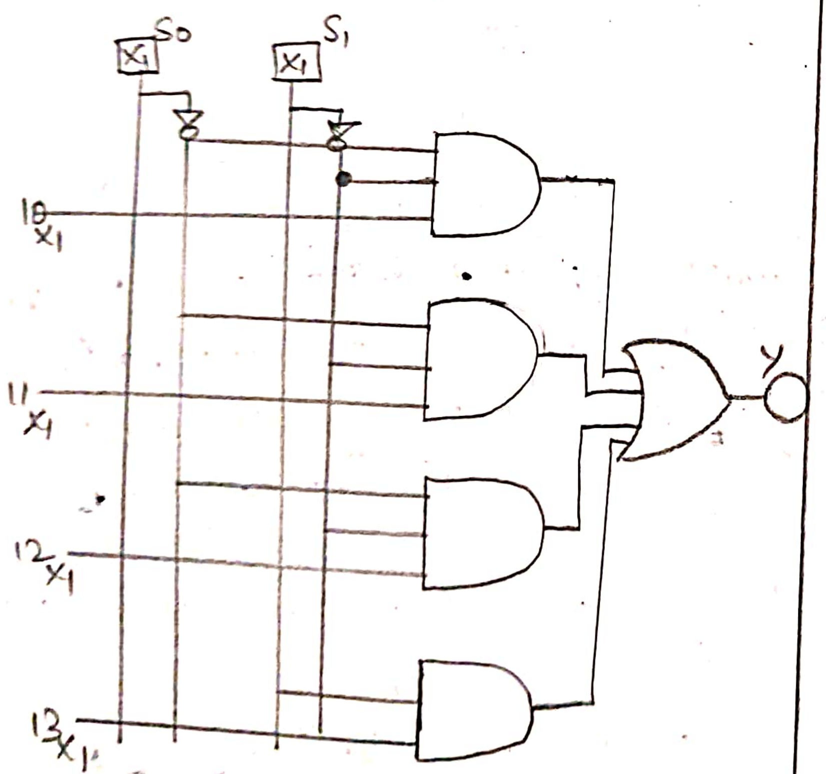
$Y_1 = S_0 S_1 I$

$Y_2 = S_0 S_1 I$

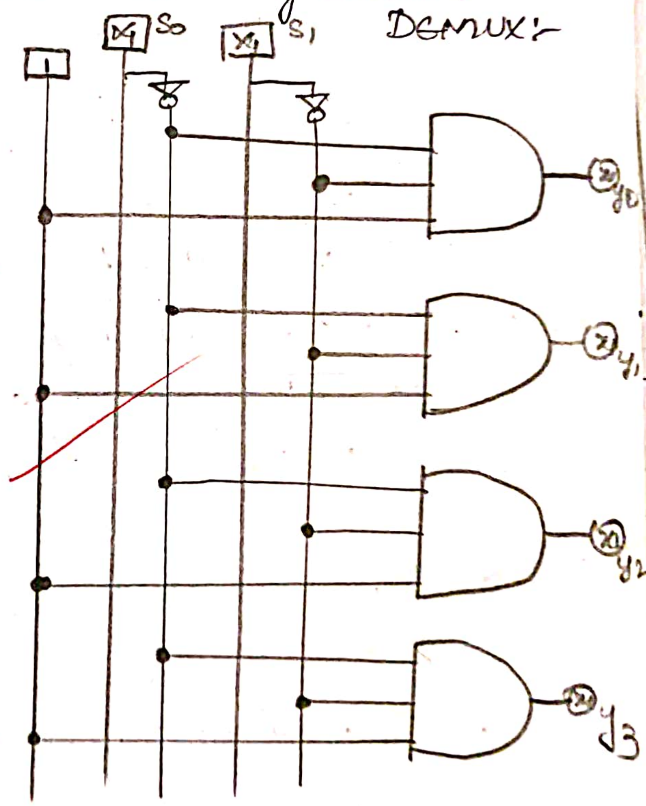
$Y_3 = S_0 S_1 I$

S_0	S_1	Y_0	Y_1	Y_2	Y_3
0	0	I	0	0	0
0	1	0	I	0	0
1	0	0	0	I	0
1	1	0	0	0	I

Internal Diagram 4:1 mux:



Internal Diagram 1:4 DEMUX:-



Conclusion : We have successfully studied and implemented 4:1 multiplexer 1:4 Demultiplexer using logisim software.

Question and Answer

1. What are different types of mux?
   Digital multiplexer. analog multiplizer; configuration based (2 to 4Omux;4 to 1 mux, 8 to 1 mux, 16 to 1mux)

2. A list out applications of mux & explain any one in detail?
   Data routing, data selection, analog to digital conversion address decoding, logic, function etc, In Communication System, multiplexer are crucial for transmission, communication System multiplexer are transmitting multiple data stores over a single physical medicen.

Experiment -07

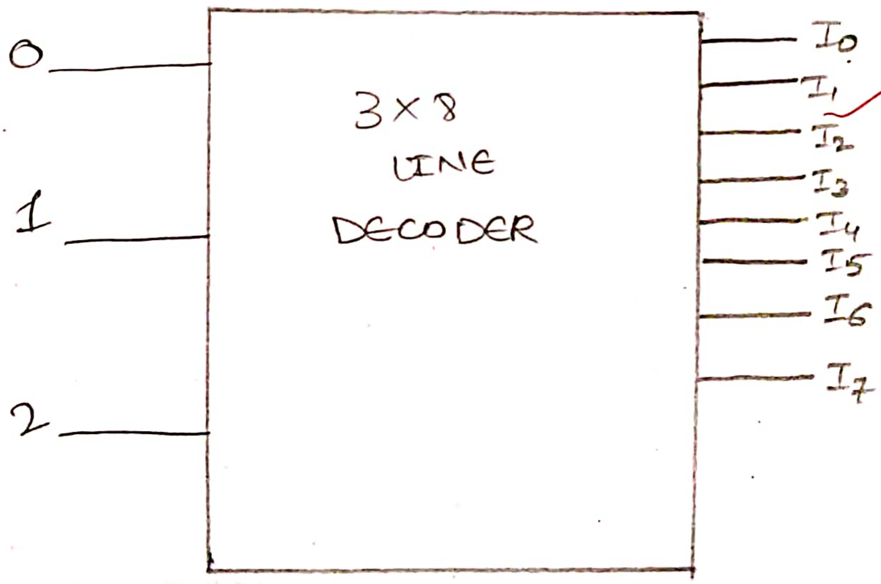
Aim: To study & design decoder circuit.

Apparatus: logic software.

Theory:

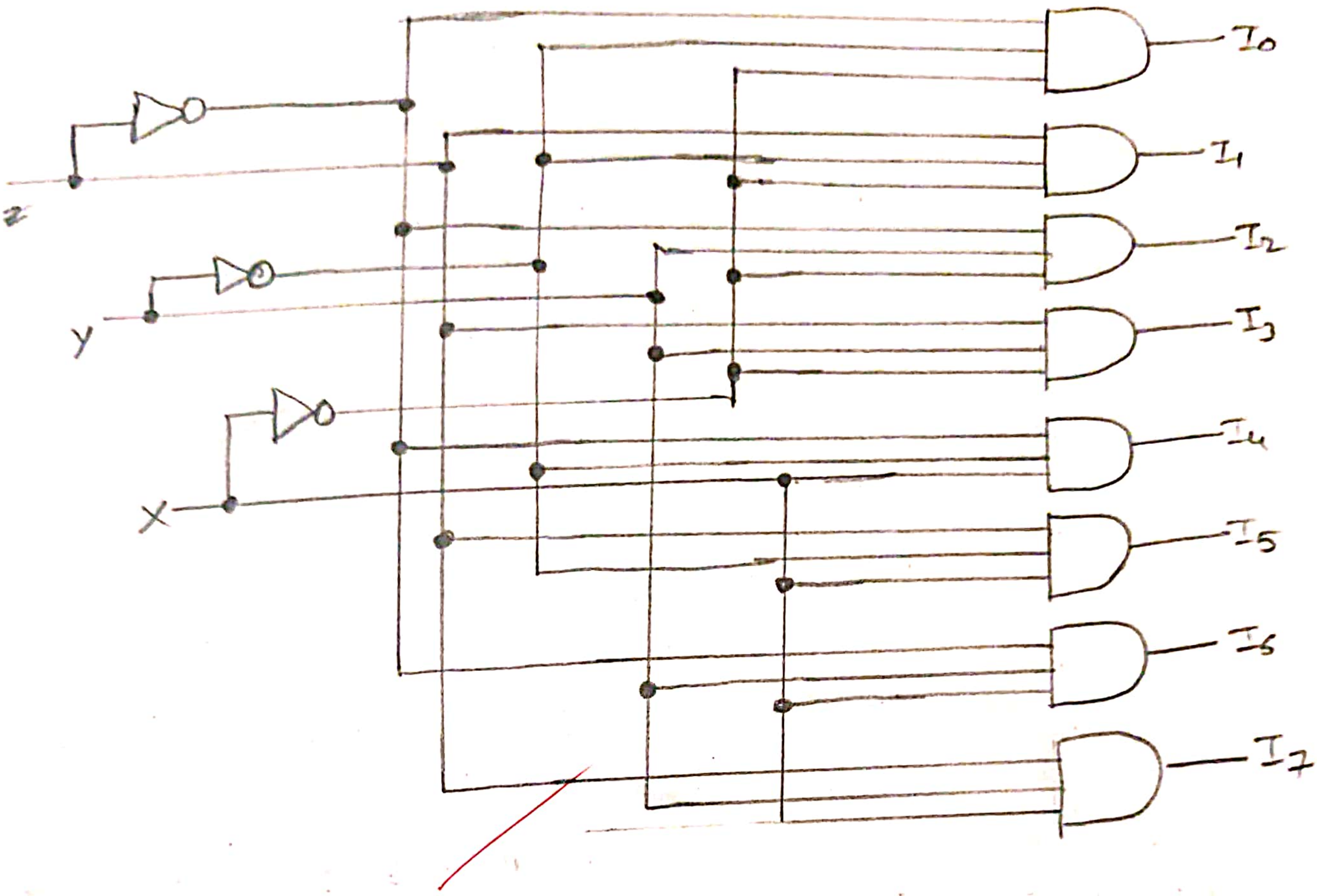
Decoder: It is a combinational circuit that converts binary information from n. input lines to a maximum. of 2^n unique o/p lines. If the n-bit decode information has unused or don't care combinations. the decoder o/p will have less than 2^n o/ps.

Block diagram:-



Circuit Diagram:

3:8 Decoder:-



Truth Table:-

Inputs			Outputs							
X	Y	Z	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Conclusion :- It's helps understand how binary inputs are converted into unique outputs. It's essential for digital systems like memory addressing, datarouting, and control applications.

Questions:-

Q1. Types of decoder circuits & 2:4 decoder explanation.

→ Types : 2 to 4 decoder
3 to 8 decoder
4 to 16 decoder
BCD to decimal decoder
7 - Segment decoder

2:4 Decoder : It has 2 input lines and 4 output lines only one output is active at a time. depending on the input combination. Ex:- if inputs are 10, output Y_2 is active.

2. Application of decoder.
→ Application: Used in memory address decoding in computers the decoder selects a particular memory location by activating only one output line corresponding to the given address input.

Q3. Make 2:4 decoder using 1:4 DEMUX.
→ A 1:4 demux can act as a 2:4 decoder if we keep the data input D=1.

Experiment : 08

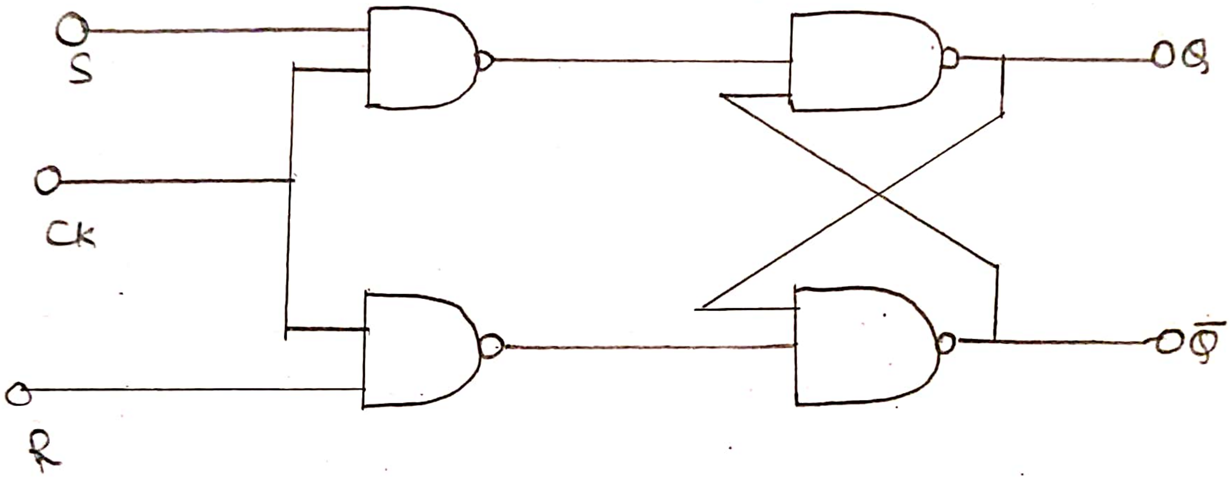
Aim:- To study and design SR Flip-Flop and D Flip-Flop.

Apparatus: Logisim Software.

Theory: Flip flop are sequential which store 1 bit of data at time. It is used to build the counter and shift register in digital circuits.

A. SR Flip-Flop

- Logic diagram



- Truth Table:

Clk	S	R	Q(n+1)
0	0	0	0
1	0	0	Hold
1	0	1	1
1	1	0	0
1	1	1	Invalid

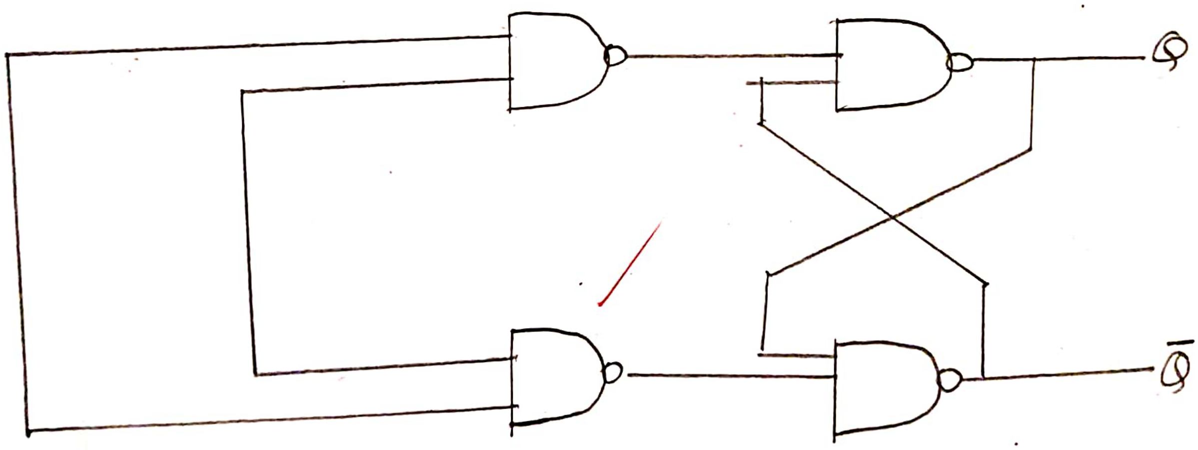
• Characteristics Table:

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	Hold
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

(B) D-Flip-Flop

The RS-FF can be modified to prevent the Race around state mentioned above by introducting an interesting for bidden gate in one of the inputs as shown fig!

• Logic diagram:



Enrollment No :

Clk	D	Q_n
1	0	0
1	1	1

• Characteristics table

Q_n	Q_{n+1}	D
0	0	0
0	1	0
1	0	1
1	1	1

• Conclusion:

The study and design of SR and D-Flip Flop help in understanding the basic building blockes of sequential circuits. They are essential for storing binary data, Synchonizing signals, and designing memory elements in digital System.

• Questions :-

1. What is the limitation of Rs Flip-Flop?

→ The main limitations of an SR-Flip-flop is the invalid state that occurs when both inputs $S=1$ & $R=1$ Simultaneously. In this case the output becomes unpredictable, making the circuit unreliable for practical use.

Q2. why clock is required?

A clock is required in Flip-Flops to synchronize the changes in output with specific time intervals It ensures stable, predictable operations by preventing random or unwanted state changes.

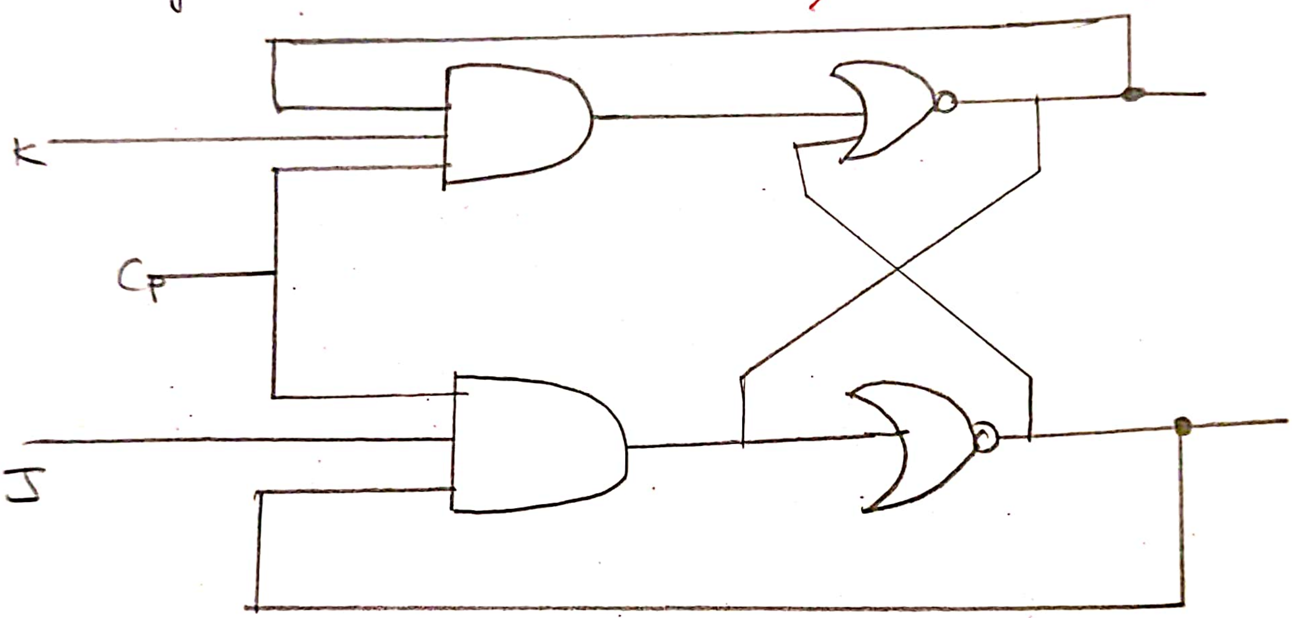
Experiment-09

Aim:- To study and verify JK Flip-Flop & T Flip Flop.

Apparatus: logism Software.

Theory: Flip-flop are sequential circuits which stores 1-bit of data at a time. It is used the counter and shifts the register in digital circuit.

Circuit diagram:



• Truth Table

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	Q_n

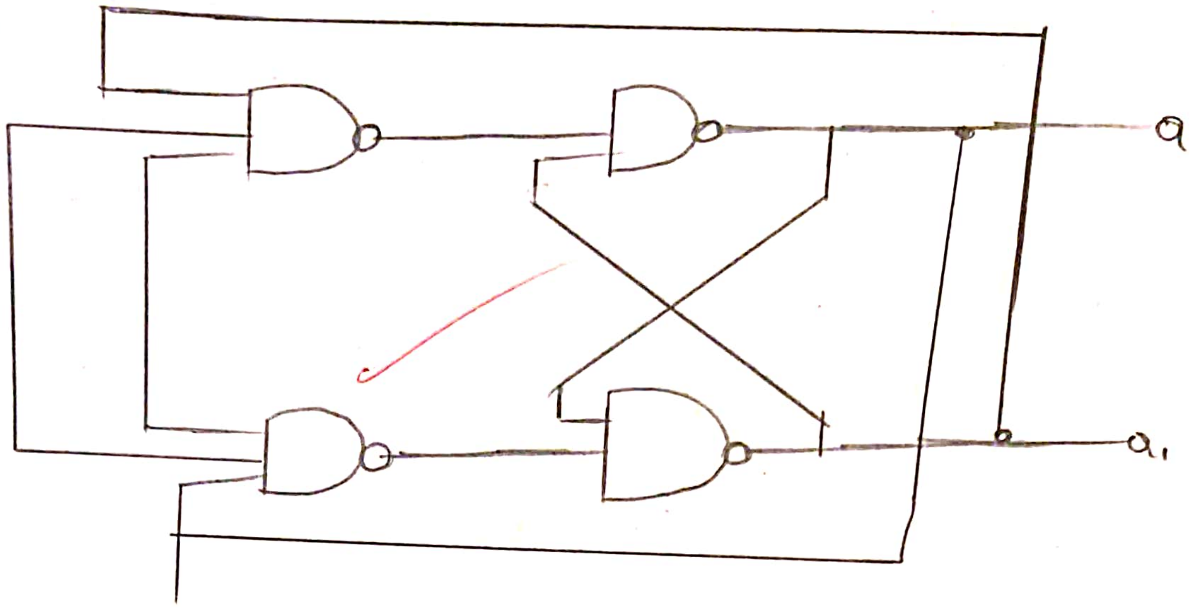
- Characteristics table:

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

$Q_{n+1} = J\bar{Q}_n + KQ_n$

- Flip-Flop:

It is possible to achieve toggling with the help of JK flip flop by simply connecting J & K inputs to high level such type of flip-flop is known as T - Flip-Flop.



• Truth Table :

T	Q_{n+1}
0	$\longrightarrow Q_n$
1	$\longrightarrow \overline{Q_n}$

• Characteristics table

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Conclusion :- The study and varification of Jk and T Flip-flop conclude that they are versatile sequential circuits used for storing one bit of data Jk Flip-Flop elimates the invalid state of SR Flip-flop while T Flip-flop is useful for counters and frequency discision.

· Questions :-

① what is the difference between JK & T Flip-Flop?

→ The JK Flip Flop has two input (J&K) and can perform set, Reset and Toggle operations while The T Flip-Flop is a simplified version with a single input (T) that only toggle the output. where T=1

② In T Flip-Flop. what is meaning of T?

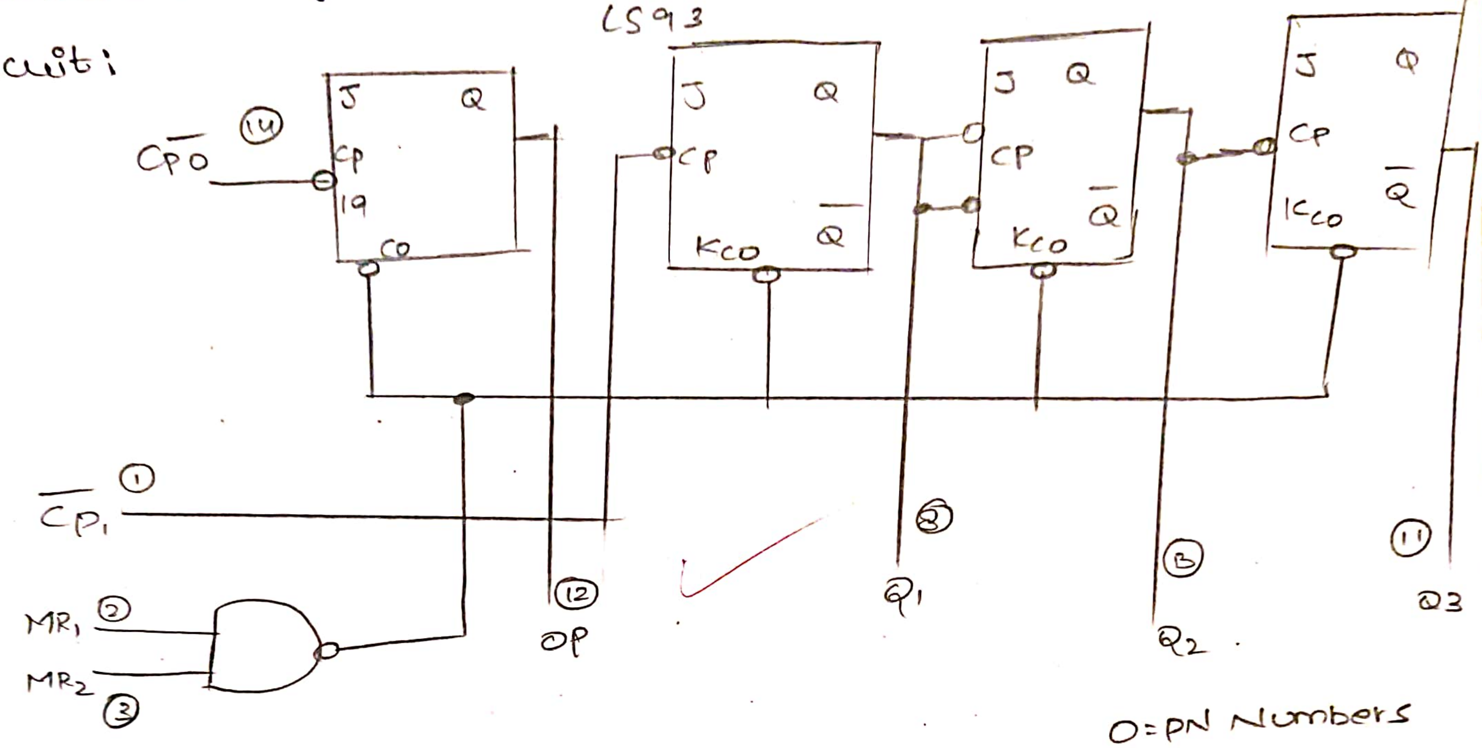
In a T Flip-Flop, 'T' stands for Toggle It means that when T=1. The output switches. from 0 to 1 or 1 to 0 on each clock pulse and when T=0 the output remains unchanged.

Experiment -10

Aim: To realize configure 4-bit ripple counter.

Apparatus : logism software.

Circuit:



O = PN Numbers
Vcc = PN5
GND = PN10

Theory:

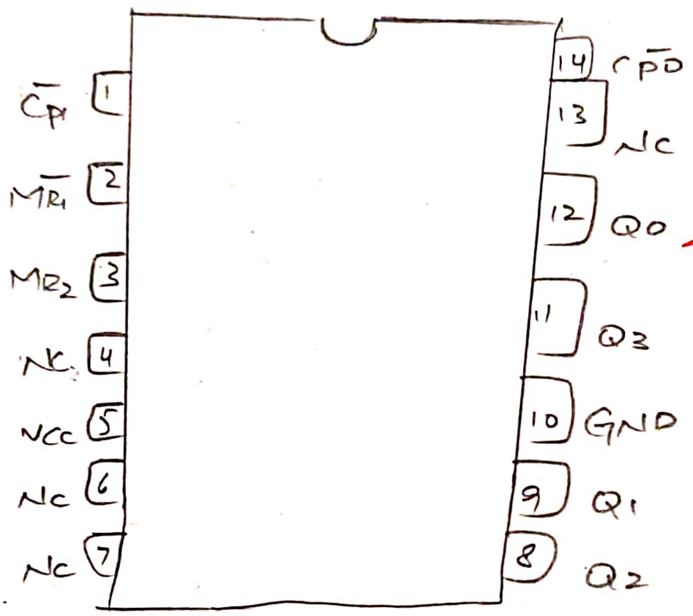
A counter is one of important digital electronic circuits asynchronous Here, we study the ripple counter or asynchronous counter are of two types which are synchronous counter & asynchronous counter. Ripple counter are constructed by using J.K flip-flop with J = 1 & K = 1. 4 J-K flip-flop are required for 4 bit counter. A square wave drives the flip-flop. The out of flip-flop a drive B flip-flop.

derive C flip-flop which derives D flip flop.
All the input J-k flip-flop are tied to +Vce. Under these situations each flip-flop changes. Its state when a positive edge clock pulse occure. This 4 bit Counter is also known as module 16 counters.

Count	2^0	2^1	2^2	2^3
1	0	0	0	0
2	1	0	0	0
3	0	1	0	0
4	1	1	0	0
5	0	0	1	0
6	1	0	1	0
7	0	1	1	0
8	1	1	1	0
9	0	0	0	1
10	1	0	0	1
11	0	1	0	1
12	1	1	0	1
13	0	0	1	1
14	1	0	1	1
15	0	1	1	1
16	1			

Pin Diagram:-

CONNECTION DIAGRAM
DIP (TOP VIEW)



NC= No internal connection

Procedure:-

* Connect +5v DC supply
* Connect pulse input to clock 1 input of IC 74C93
* Connect Qa, Qb, Qc & Qd output to D1, D2, D3 & D4 inputs of BCD display.
* Connect Jumper Between the Clk2 & Q4 output.
* Connect either Pin 2 (or) pin 3 (rest output) to ground.

* Apply Counting pube by pushing Switch Apply one plus at a time. The BCD Show 0 to 9 Counting & then will show five other State after 10th pulse then after 16th pulse output will reset to zero.
* You can also Supply continue to pulse by connecting clock generator output pulse.
* The output, Qa, Qb, Qc and Qd Can also be Connected output indicators to observe binary output

Conclusion:-

The 4 Bit ripple Counter was Successfully built and tested using J-K flip-flop it worked as asynchronus Counter where each flip flop It worked as an state on the positive edge of the clock pulse the outputs correctly Showed the Binary Counting Sequence from 0 to 15 the Connection diagram and truth table where checked during the experiment the Results Matched the Expected working principle of ripple Counter.

Questions:-

1) List out of different types of counter.

- Asynchronous Counter:- flip-flop are not clocked together; the output of one flip-flop out as the clock next
- Synchronous Counter:- All flip-flops are clocked at the same time using common clock
- Decade Counter:- These count from 0 to 9 and then reset (Module -10)
- up/Down Counters:- Can count both upward & downward (n, n-1, n-2)
- Ring Counters:- A Shift Register where the output of the last flip-flop is connected back to the input of the first forming Ring.
- Johnson-Counter:- A Modified ring counter where the inverted output of the last flip-flop is fed back to the input of the first.

2Q) What is Modulo counter

A Modulo Counter (Modulo- N-Counter) is a digital Counter that counter up to a spectic Number N and then resets Back to zero the Number N shows the total Number of distinc States that counter can go through Before Repeating.